Pedestrian Detection for Collision Avoidance in Driving (5m-30m)

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***Abstract*—The purpose of this research is to analyze the efficiency of YOLOv8 for pedestrian detection in collision avoidance between 5m-30m and compare it with SSD. The demand in ADAS is increasing lately and pedestrians are most vulnerable in the traffic.[8] This research evaluates detection accuracy and inference speed to compare YOLOv8 with SSD models between 5m and 30m. Preliminary observations indicate YOLOv8's performance supasses SSD model in terms of accuracy and a higher mAP score.[1] Thus, YOLOv8 is a suitable candidate for avoid collision. This paper also discusses the efficiency between YOLOv8 and the latest YOLOv11n models to apply in the mid-range pedestrian detection module for real-time driving assistance systems. As a result, YOLOv8 is suitable for deployment on edge devices because of the speed and lightweight model. YOLOv11n provides improvement in accuracy than YOLOv8n. Therefore,it's better to apply for the systems with abundant resources.**

I. INTRODUCTION

According to statistical reports on traffic accidents. Human factors are the main reason to lead to accidents. For instance, driver distraction, drunk driving, speeding, fatigue, and running red lights are a few of the most common causes of serious injury accidents. Pedestrians are the weakest level in the traffic pyramid which means they are the highest priority to be protected on the road. According to the World Health Organization (2023), over 270,000 pedestrians are killed each year in road traffic crashes worldwide, accounting for 23% of all traffic-related deaths. Many of these fatalities occur in urban areas at intermediate distances (5–30 meters) Pedestrian detection plays a significant role in transport systems, autonomous vehicles for safety systems.[7][9]

As Sergiu Nedevschi *et al.* [6] proposed, stereo vision systems were used in early pedestrian detection tasks for collision avoidance. R,Aufrère et al. [9] proposed a multi-sensor perception system for autonomous vehicles that combines video, laser rangefinders, and optical flow to detect and predict potential collisions, particularly with pedestrians. Their work focuses on 360-degree short-range sensing, which is similar to the core objective of this project: pedestrian detection for forward and reverse collision avoidance.

Traditional object detection algorithms such as Haar cascades and HOG+SVM have shown limitations in accuracy and adaptability under diverse lighting and environmental conditions.[12] In contrast, deep learning-based models—particularly those in the YOLO (You Only Look Once) family—have become the standard model due to their balance between speed and accuracy .

While numerous studies have focused on the deep learning model such as YOLO, Faster R-CNN and SSD in object detection.[3][4] However, their detections are based on different environment and distance conditions. In summary, This study aims to compare YOLOv8n and YOLOv11n in three main aspects:

1.It focuses on pedestrian detection and avoiding collisions within mid-range(5–30 meters) in the requirements of real-time embedded driving systems.

2.It provides a comparative study between YOLOv8n and the latest YOLOv11n model, focus on the trade-off between model accuracy and computational efficiency.

3.It integrates both quantitative and qualitative evaluations, including PR/F1 curves and visual detection results to compare the performance with test results.

II.BACKGROUND

Pedestrian detection is a branch of computer vision which is commonly used in the field of autonomous driving, monitoring systems and collision prevention. There are many detection models that come up with the development of deep learning like R-CNN,SSD, YOLO and so on. Among these, YOLO became a popular model because of the advantages in real-time and balance between accuracy and speed.[5] The structure of YOLOv8 adopts anchor-free and Ultralytics tools to train. Therefore,YOLOv8 is flexible for developers to train, validate and deploy which is convenient to implement.

However, most datasets and pre-trained models focus on general objects with long ranges. In contrast, these models aren't so suitable for pedestrian detection between 5m and 30m. Although SSD and Faster R-CNN models provide excellent accuracy,their inference time and set-up complexity is high which is disadvantageous in real-time embedded systems.[2] Therefore, this research aims to train a custom pedestrian detection model by YOLOv8 and medium-range distance images as a source to specially focus on pedestrian vs. non-pedestrian binary classification tasks. The goal in research is to optimize the model for detection accuracy and inference accuracy in medium-range distance which is for real-time ability to react to real-world situations.

III. METHOD

In this research,we focus on three algorithms of computer vision for pedestrian detection and collision avoidance. We selected **MOG2**, **YOLOv8n and YOLOv11n** to compare the performance in pedestrian detection by different perspectives .

A.**MOG2(Mixture of Gaussians v2)**

MOG2 offers background modeling and foreground objects based on OpenCV by using the GMM model to dynamically learn the background in image[11]. The advantages are suitable in fixed-camera scenes to real-time detect the pedestrian's contour. MOG2 can provide for understanding visual behaviors in foreground masks under varying lighting and shadow situations although it can't offer class information and precise boundaries in images.

B.**YOLOv8n**

YOLOv8n is an Ultralytics YOLO model. The YOLO series models have the advantages for real-time scene detection because they have high efficiency and small modules by optimizing the low resource devices. This study already trained the pedestrian detection dataset by the YOLOv8n model and gained remarkable performance in 0.92 mAP50 and 0.71 mAP50-95. YOLOv8 n can perform bounding box regression and class classification in single forward pass in excellent inference speed with end-to-end architecture.

The entire diagram of this study is summarized in Figure 1. It shows the step-by-step pipeline for model training and inference using the Ultralytics framework, from dataset preparation and annotation to evaluation and visualization.[13]

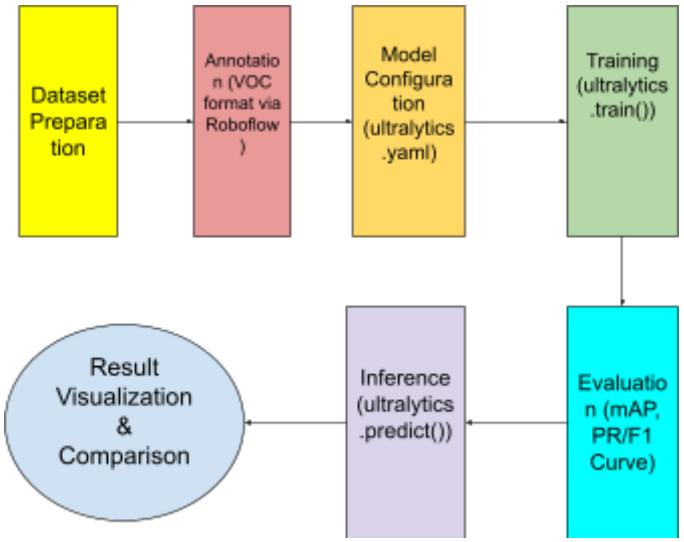

Figure 1:Overview of the YOLOv8n and YOLOv11n model pipeline implemented in this paper, including dataset preparation, annotation, training via Ultralytics framework, evaluation, and result comparison.

C.**YOLOv11n**

YOLOv11n is the latest version in YOLO architecture which improves the training steps, model architecture and strategy for data augmentation. It supports AMP, automatic batch size adjustment and auto-anchor learning to achieve higher accuracy in a shorter time. YOLOv11n adopts advanced attention modules and module design for pedestrian detection which shows high potential in pedestrian detection.[13] We trained the dataset by using YOLOv11n, aiming to compare the performance between YOLOv8n in this research.

D.Evaluation Metrics

Intersection over Union (IoU) is used to quantify the overlap between the predicted and ground-truth bounding boxes. It serves as the foundation for computing mAP metrics. In this study, we report both mAP@0.5 and mAP@0.5–0.95 to evaluate detection accuracy[10]

$$IoU = \frac{Area\ of\ Overlap}{Area\ of\ Union} \quad (1)$$

The **Precision-Recall (PR) curve** illustrates the trade-off between precision and recall in different confidence situations. The area under the PR curve indicates balance between detecting true positives from minimizing false positives. The curve is used to evaluate models of unbalanced datasets like pedestrian detection when the objects in the background obviously are over the pedestrians.[15]

$$Precision = \frac{TP}{TP+FP} \quad (2)$$

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

The **F1 score** combines precision and recall into a single indication by taking their harmonic mean. It provides a more balanced view of the model's performance when one metric is disproportionately higher than the other. F1 score implies that the model has strong precision and recall which is reliable for real-world deployment scenarios in false alarms and missed detections.[16]

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (4)$$

This study can represent the advantages and disadvantages in different generations of technology in pedestrian detection tasks through the comparison with three methods we mentioned above. Meanwhile, they can provide applications for future real-world traffic applications and embedding systems.

The **overall loss function** in YOLOv8 is sum of three components: classification loss L_{cls} ,objectness loss L_{obj} and bounding box regression loss L_{box}.[13]

$$L = \lambda_{cls} \cdot L_{cls} + \lambda_{obj} \cdot L_{obj} + \lambda_{box} \cdot L_{obj} \quad (5)$$

λ_{cls},λ_{obj},λ_{box} are used to tune hyperparameters to balance each component's contribution to the total loss. The classification loss measures the cross-entropy[13] between predicted and true class labels. The objectness loss evaluates the confidence that an object exists in the predicted box. The box of loss handles the spatial accuracy in predicted bounding boxes.[13]

IV. RESULTS

Device:

- **GPU: NVIDIA GeForce RTX 4060**
- **OS: Ubuntu 22.04 LTS**
- **Processor:Intel(R) Core(TM) i7-14700**
- **Memory: 32GB RAM**
- **IDE: NeoVim**
- **Programming Language: Python 3.12.3**

Camera:

- **Dataset Source: Roboflow Pedestrian Detection Dataset**
- **Resolution: 1080p**
- **Frame Rate: 30 FPS (simulated from dataset)**

Software:

- **Ultralytics YOLO Version: 8.3.126**
- **PyTorch Version: 2.7.0 + CUDA 12.6**
- **OpenCV Version: 4.9.0**
- **Numpy Version: 1.26.4**

This research uses Pedestrian Detection v1i dataset from Roboflow[14] by fixed validation split to train the model in 50 epochs for evaluation performance in the pedestrian detection algorithm. We will compare such as mAP,precision and recall which are recorded in three algorithms.

A.YOLOv8n performance

YOLOv8n is a modern and lightweight model with about 3 million parameters, showing the excellent inference speed and efficiency coverage. The training result is below.

Table I
YOLOv8n performance in our training

mAP50	0.92
mAP50–95	0.71
Precision	0.93
Recall	0.83

YOLOv8n performance on the pedestrian detection task using the custom dataset. The indicators(mAP50 & mAP50-95) show that YOLOv8n is highly suitable in real-time pedestrian detection,especially for the embedded system which has resource restriction.

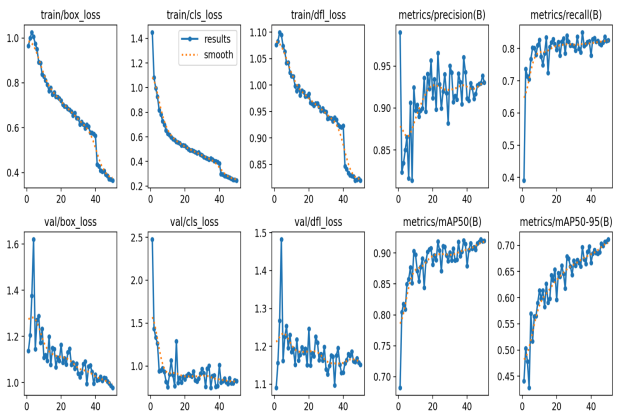

Figure 2. Training and validation curves for YOLOv8n. These include training losses (box, classification, and DFL), evaluation metrics (precision and recall), validation losses and performance metrics (mAP@0.5 and mAP@0.5:0.95). Overall, the model exhibits stable convergence and consistent performance improvement in 50 epochs.

B.YOLOv11n performance

YOLOv11n is the latest model in the YOLO series, implementing advanced

architectural enhancements, including attention modules, automatic anchor learning, and AMP (mixed precision training). Its training results are as follows:

Table II
YOLOv11n performance in our training

mAP50	0.94
mAP50–95	0.74
Precision	0.95
Recall	0.85

The result shows YOLOv11n has better performance than YOLOv8n,especially for recognizing smaller or occluded pedestrians. However, YOLOv11n takes longer to train because of its complexity.

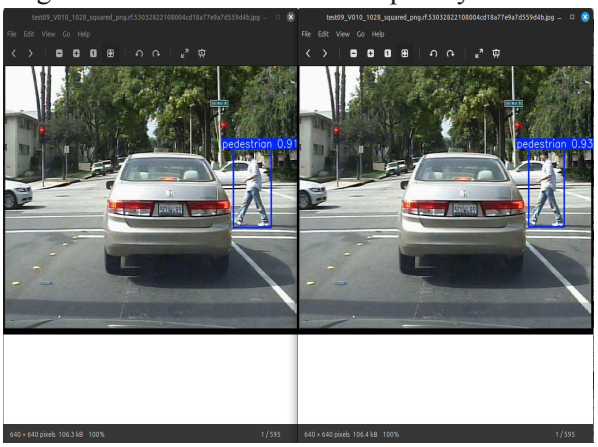

Figure 3. Left side is the result from YOLOv8n, the right side is from YOLOv11n.

The scene is of an occluded pedestrian by a vehicle. YOLOv8n gained 0.91 confidence from this scene. However,YOLOv11n has higher 0.93 confidence than YOLOv8n. This result illustrates YOLOv11n is suitable in complex scenes or occluded situations.

C. MOG2 performance

MOG2 is demonstrated by OpenCV to evaluate. MOG2 effectively highlights foreground motion regions in a fixed camera viewpoint. It can outline pedestrian contour accurately with a stable lighting scene but the accuracy will decrease in situations with dynamic backgrounds and significant lighting changes.[11]

D.**Model Comparision**

Table III

model	mAP50	mAP50-95	Precision	Recall
MOG2	N/A	N/A	N/A	N/A
YOLOv8n	0.92	0.71	0.93	0.83
Yolov11n	0.94	0.74	0.95	0.85

YOLOv11n achieved better performance in all evaluation indicators.

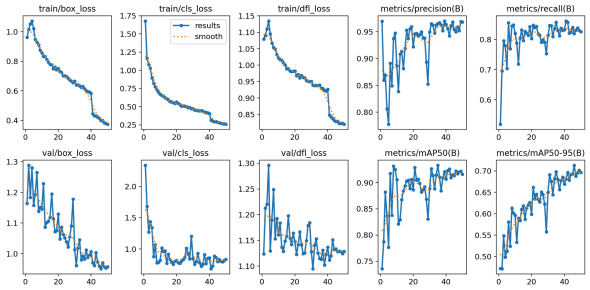

Figure 4. YOLOv11n training and validation metrics. The first row shows training losses and evaluation metrics. The second row shows validation losses and mean average precision scores. All metrics are recorded over 50 epochs.

Here are some of the predicted results between YOLOv8n and YOLOv11n. The left side displays the predictions from YOLOv8n, the right side represents the prediction

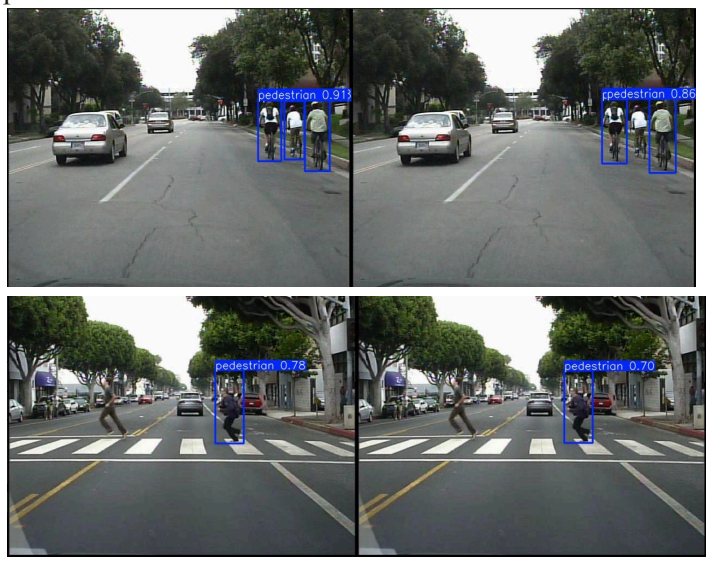

Figure 5: Comparison of predicted bounding boxes between YOLOv8n (left) and YOLOv11n (right) on six test scene

YOLOv11n predicts more precise bounding boxes and better localization, especially in occlusion and cluttered backgrounds. The visual results provide the conclusion that even though YOLOv11n is slower and heavier to train, it's still a stronger model for mid-range pedestrian detection in real-world driving situations.

Comparison Precision-Recall Curve and F1 Score between YOLOv8n and YOLOv11n. The PR/F1 curve shows the advantages of YOLOv11n in accuracy and stability.

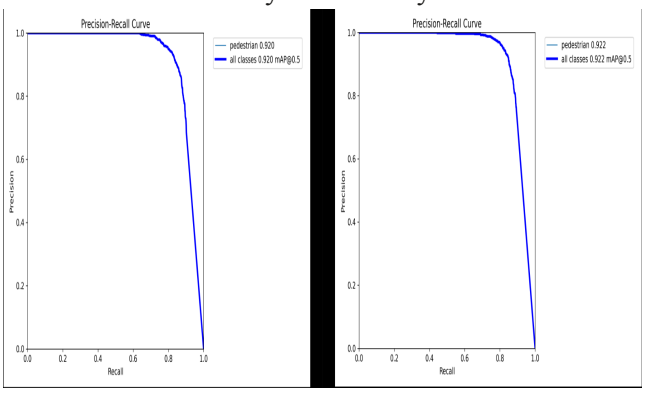

Figure 6. Precision-Recall curve of YOLOv8n(left) and YOLOv11n(right).

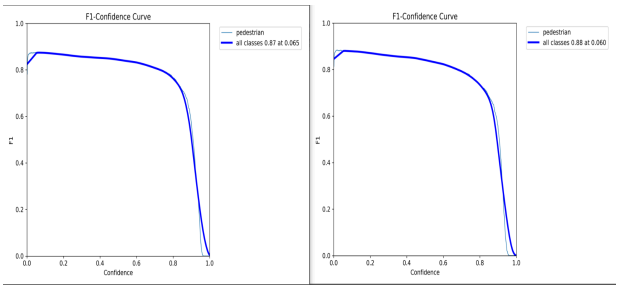

Figure 7. F1-score curve showing performance variation during training.

V. CONCLUSION

This research presents three representative algorithms in computer vision for pedestrian detection,such as the traditional algorithm MOG2,modern lightweight YOLOv8n and next-generation model YOLOv11n. These experiment results reveal their advantages and disadvantages in special situations.

MOG2 can't provide class information or bounding box information but it offers inexpensive foreground object detection in static-camera scenarios. YOLOv8n has excellent performance in a real-time embedded system because of its efficient architecture and 3 million parameters. YOLOv11n improves accuracy by using attention modules and automatic anchor learning which is suitable for small or occluded pedestrians.
YOLOv11n has better performance than YOLOv8n in quantitative results(mAP50:0.94 vs. 0.92,Precision:0.95 vs. 0.93) and stable inference speed.

In conclusion, YOLOv8n has a strong balance between speed and accuracy. Therefore, it's ideal for a real-time scene. YOLOv11n is suitable for the accuracy priority situation. This study offers practical insight for selecting appropriate algorithms in pedestrian detection systems,especially for collision avoidance applications

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